# MNDAF-Net: Multiscale Neighborhood Decoupling and Attribute-Focused Fusion for Dual-Temporal Remote Sensing Image Scene Change Detection Network

Zhengpeng Li, *Graduate Student Member, IEEE*, Jun Hu, *Member, IEEE*, Bin Yang, Kunyang Wu, and Jiansheng Wu

***Abstract*—Existing methods in remote sensing image scene change detection typically rely on convolutional or Transformer architectures to extract attribute features, combining local details with global semantic modeling. However, these approaches often overlook the importance of neighborhood attribute features for capturing local changes and contextual stability, leading to insufficient representation of fine-grained changes. In addition, bitemporal feature fusion commonly suffers from information loss, limiting the depth of interaction modeling in change regions and impacting both accuracy and robustness. To address these challenges, we propose a novel dual-temporal remote sensing scene change understanding network with multiscale neighborhood decoupling and attribute-focused fusion. In the encoding phase, we leverage a multiscale neighborhood feature decoupling analysis (NFDA) module to enhance independent modeling of neighborhood attributes, effectively capturing fine-grained changes in complex scenes. Specifically, this module integrates multihead adjacent atrous attention ($MA^2$) and a panoramic adaptive attribute sensing ($PA^2S$) layer to provide comprehensive multiscale perception and adaptive feature enhancement for bitemporal remote sensing images. In the decoding phase, the attribute-focused fusion decoder (AFF-Decoder) enhances the fusion of bitemporal features by ensuring correlated interactions between dual-temporal features, capturing fine-grained relationships to achieve more accurate change representations. Experimental results on three public datasets—LEVIR-CD, WHU-CD, and DSIFN-CD—demonstrate that the proposed MNDAF-Net achieves promising performance in remote sensing attribute change understanding. Our code will be published at https://github.com/lzp-lkd/MNDAF-Net***

Received 5 April 2025; revised 15 June 2025 and 13 July 2025; accepted 31 August 2025. Date of publication 3 September 2025; date of current version 9 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62376036, in part by the Fundamental Research Project of Higher Education Institutions by Liaoning Provincial Department of Education under Grant LJ222410146057 and Grant LJ212410146040, and in part by the Graduate Education Reform and Scientific and Technological Innovation and Entrepreneurship Project of the University of Science and Technology Liaoning under Grant LKDYC202403. *(Corresponding author: Jun Hu.)*

Zhengpeng Li, Jun Hu, Bin Yang, and Jiansheng Wu are with the University of Science and Technology Liaoning, Anshan 114051, China (e-mail: lkdlzp0901@163.com; 320083700074@ustl.edu.cn; 320193700107@ustl.edu.cn; ssewu@163.com).

Kunyang Wu is with the Key Laboratory of Geophysical Exploration Equipment, College of Instrument Science and Electrical Engineering, Jilin University, Changchun 130061, China (e-mail: wuky23@mails.jlu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3605637

***Index Terms*—Attribute change understanding, attribute compression (AC), feature fusion, multiscale feature decoupling, neighborhood feature analysis.**

## I. Introduction

ATTRIBUTE change detection in remote sensing imagery aims to accurately capture land cover changes in the same area over different time periods, providing critical support for urban management [1], [2], environmental monitoring [3], and disaster assessment [4]. However, this task faces challenges such as diverse data sources, complex land cover types, and varying illumination conditions [5]. These factors collectively increase the complexity of attribute change understanding, necessitating algorithms that not only detect subtle changes but also distinguish real changes from pseudo-changes induced by factors such as lighting and seasonal variations, thereby enhancing interpretative robustness [6].

Traditional approaches to attribute change understanding include algebraic, transformation, classification, statistical, and object-based methods. Algebraic methods are computationally simple but prone to noise interference [7], and transformation methods emphasize changes through feature space mapping but often perform poorly in complex scenes [8], [9], [10]. Classification methods depend on the quality of training samples, while clustering approaches struggle to capture change characteristics in intricate backgrounds [11], [12], [13], [14]. Object-based methods can incorporate spatial information but are highly dependent on segmentation quality [15], [16]. Overall, these approaches struggle to model interfeature relationships effectively in complex multitemporal remote sensing images and lack robustness in handling background complexity, limiting detection accuracy. Fig. 1 illustrates changes in land cover between two time points (T1 and T2), highlighting significant differences in the shape, shadow, and position of buildings. These subtle changes are challenging for traditional methods to capture accurately, often resulting in increased false changes. The "postchange attribute" section at the bottom of the figure further details the distinctive attribute characteristics of altered features, underscoring the critical need

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

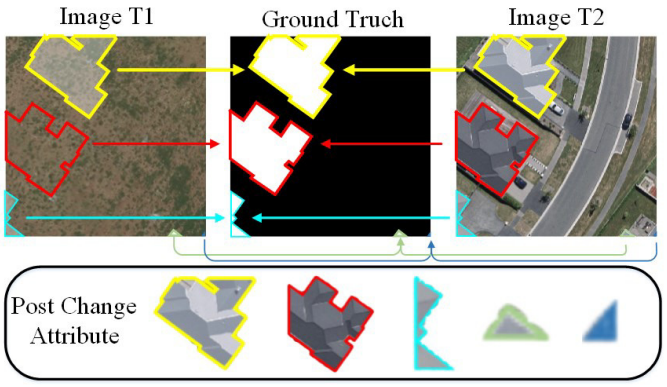



Fig. 1. Example of land cover changes in multitemporal remote sensing images and postchange attribute characteristics.

for fine-grained change detection and attribute recognition in complex scenes.

With the advancement of deep learning, particularly the introduction of CNNs, attribute change understanding methods have made significant strides. Early deep learning models employed feature extractors to input bitemporal features into classification networks, achieving some success. Contemporary models increasingly adopt Siamese fully convolutional networks, which use feature learning to better represent complex change areas [17], [18]. However, these networks primarily focus on extracting and fusing global and detailed features, often overlooking the implicit contextual relationships and connections between neighboring attributes within bitemporal features. This can limit the accuracy of identifying localized changes. In addition, feature fusion processes are prone to information loss, lacking the capacity to fully capture the fine-grained interactions between change regions, thereby impacting detection accuracy and robustness. These limitations have driven research toward refined neighborhood feature modeling and decoupling methods to capture more distinctive change features at both global and local scales.

More recently, the application of Transformer architectures in visual tasks has grown, with their self-attention mechanisms excelling at modeling long-range dependencies. By integrating attention mechanisms with CNNs, researchers have achieved notable progress in modeling long-range contextual information [19], [20]. For example, Transformer variants like ViT demonstrate advantages in multiscale feature fusion [21], [22], [23], [24]. Traditional attention mechanisms, however, often focus on global feature relationships, typically overlooking associations among neighboring pixels in space. In complex scenarios involving multiattribute interactions, the lack of neighborhood information can lead to dilution or loss of local features. To address this, we propose the multihead adjacent atrous attention ($MA^2$) based on recent advancements in Transformer research, aiming to optimize bitemporal feature alignment and improve representation of change areas.

Traditional attribute change understanding approaches also face numerous challenges in bitemporal feature fusion. First, these methods often encounter issues with information redundancy and feature loss in change region representations [25], [26], a challenge particularly evident in multiscale and complex scenarios. Second, feature fusion techniques frequently rely on simple stacking or concatenation, lacking deep exploration of fine-grained semantic information, which results in suboptimal detection accuracy for change regions in complex settings [27]. While recent approaches combining CNNs and Transformers have enhanced attribute change understanding through improved long-range contextual modeling, they still inadequately address local attribute associations among adjacent pixels, limiting the stability of detection outcomes in complex scenarios.

To address these limitations, this article proposes an MNDAF-Net. This approach introduces an $MA^2$ mechanism and a panoramic adaptive attribute sensing ($PA^2S$) layer to enhance the capture of local and global change information through multiscale neighborhood feature decoupling, effectively addressing deficiencies in neighborhood information modeling in traditional methods. In addition, we designed an attribute-focused fusion decoder (AFF-Decoder) to condense bitemporal features into essential semantic tokens, reducing redundancy and focusing on key features of change regions. To further improve feature fusion, a cross-attention mechanism is incorporated to capture fine-grained interactions within change regions, ensuring robust detection of subtle changes in complex scenarios.

The main contributions of this article are as follows.

1) We propose a multiscale neighborhood decoupling and attribute fusion framework for bitemporal remote sensing attribute change detection, enhancing the capture of fine-grained features within change regions.
2) To address the shortcomings in neighborhood attribute feature learning for attribute change understanding, we design the $MA^2$ and $PA^2S$ layer, which effectively strengthen change understanding accuracy and robustness in complex scenarios by filling gaps in neighborhood information utilization.
3) Designed an AFF-Decoder, which, through semantic compression and cross-feature learning, suppresses redundant features and focuses on the key attributes of the change area, achieving deep interaction and refined difference capture of bitemporal features.

The remainder of this article is organized as follows. Section II reviews the related work. Section III presents the detailed architecture and implementation of MNDAF-Net. Section IV describes our experimental setup. Section V analyzes the experimental results. Finally, Section VI concludes this article and discusses future research directions.

## II. Related Works

### A. CNN-Based Attribute Change Detection

CNNs are widely applied in understanding attribute changes in remote sensing images, leveraging their capacity for local feature extraction and multiscale information fusion to enhance the accuracy of change localization [28], [29], [30], [31], [32]. For example, Chen et al. [33] combined nonlocal feature pyramid networks with dense connectivity modules, improving multiscale feature extraction and fusion, though

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

the method is limited in capturing fine-grained neighborhood attributes. Tian et al. [34] enhanced spatiotemporal feature modeling by generating a saliency probability map, increasing detection accuracy; however, it still lacks depth in dual-temporal feature interactions. While these works advance change detection accuracy, challenges remain in effectively utilizing neighborhood information and capturing fine-grained features, with false positives (FPs) and misdetections persisting in complex scenes.

Recently, approaches combining CNNs and graph convolutional networks (GCNs) have gained attention as they leverage graph structures to enhance neighborhood information modeling [35], [36], [37], [38], [39]. Xiong et al. [40] proposed a wavelet-based Siamese network to reduce semantic differences between dual-temporal images. Wang et al. [41] combined CNNs and GCNs to highlight edge information, improving detection accuracy. Though these hybrid models partially address neighborhood information gaps, challenges persist in handling feature redundancy across dual-temporal images and capturing subtle changes, limiting their effectiveness in high-resolution remote sensing applications. Zhang et al. [42] integrated CNN-based local feature extraction and graph neural network (GNN)-based global feature modeling to improve attribute change understanding. Yang et al. [43] employed multilevel feature fusion and attention mechanisms to precisely delineate change boundaries. Zhang et al. [44] presented a combined CNN–GNN approach to enhance change detection in remote sensing images, while Long et al. [45] used graph learning to quantify interactions between dual-temporal semantic and difference features, improving performance in complex scenes. Tang et al. [46] introduced a multitask learning model that incorporates semantic and change relationships to enhance high-resolution attribute change understanding. Cheng et al. [47] extracted fine-grained image features through central differential convolution and used adaptive deep-shallow feature fusion to capture dependencies between deep and shallow features, further improving detection accuracy.

Some studies have innovated within the CNN framework to address change complexity and reduce reliance on annotated data. For instance, Chen et al. [48] introduced an approach that generates change regions by exchanging patches within and across images, obviating the need for dual-temporal pairs and labels, thus proposing a new approach for attribute change understanding. Zhang et al. [49] developed a globally aware Siamese network that mitigates class imbalance between change and nonchange regions by integrating global attention and foreground-awareness modules to leverage contextual and foreground–background relationships. Sun et al. [50] applied similarity relationship graphs to describe low-frequency information and used dissimilarity graphs to capture high-frequency information, enhancing attribute change detection accuracy.

While these CNN–GCN hybrid models improve detection accuracy to some extent, they remain limited in effectively using neighborhood attribute information and capturing fine-grained changes. To address these challenges, we propose MNDAF-Net, which introduces multiscale neighborhood decoupling and AFF-Decoder to adaptively capture neighborhood features and enhance the modeling of change regions in complex scenes.

### *B. Transformer-Based Attribute Change Detection*

Transformer models have recently shown great potential in understanding attribute changes in remote sensing images. Their ability to capture global context and model long-range dependencies has made them an emerging approach in this field. For example, Chen et al. [21] proposed representing spatiotemporal context information as semantic tokens, using a Transformer encoder to model relationships among these tokens, enhancing the accuracy of attribute change understanding. Bandara and Patel [51] effectively combined a hierarchical Transformer encoder with a lightweight MLP decoder to extract and fuse multiscale features for change detection in remote sensing imagery. Xu et al. [52] further improved the accuracy of attribute change understanding by integrating the advantages of CNNs and Transformers to enhance multiscale feature extraction and fusion.

Recent research has also begun exploring models that combine GNNs with Transformers to improve background modeling capabilities. Jiang et al. [22] applied this hybrid approach to emphasize unchanged background information, achieving higher precision in attribute change understanding. Tang et al. [53] developed a W-shaped hierarchical network with dual Siamese branches to simultaneously extract local fine-grained features and global contextual information for attribute change understanding in remote sensing images.

Progress has also been made with unsupervised approaches using Transformer architectures. Fang et al. [54] proposed an unsupervised attribute change understanding method based on content-invariance translation and a hybrid attention mechanism. Through adversarial learning and hybrid attention, they suppressed style differences between pretemporal and post-temporal images, enhancing robustness in change detection. In addition, Yu et al. [55] and Li et al. [56] introduced multiscale convolutional attention mechanisms and relative position encoding to further improve global feature extraction and representation of change regions, aligning with the neighborhood attribute extraction concepts proposed in our study.

For multiscale feature fusion, Jiang et al. [57] proposed a CNN-Transformer network that extracts and fuses multiscale features, leveraging the strengths of both CNNs and Transformers for attribute change understanding in high-resolution remote sensing imagery. Lei et al. [58] designed a cross-dimensional interactive self-attention module to reduce computational complexity, along with a structure-aware enhancement module to refine fine-grained feature modeling. Moreover, Liu et al. [59] combined spatial and frequency domain information, using frequency-gated attention to mitigate style differences in images, thereby improving model robustness. Hu et al. [60] introduced GlobalMind, a global multihead interactive self-attention network focused on attribute change understanding in hyperspectral images. This approach uses a global axial partitioning strategy combined with multihead interactive self-attention to capture global spatial and cross-temporal feature correlations. Qian et al. [61] proposed a temporal–spectral–semantic-aware convolutional

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

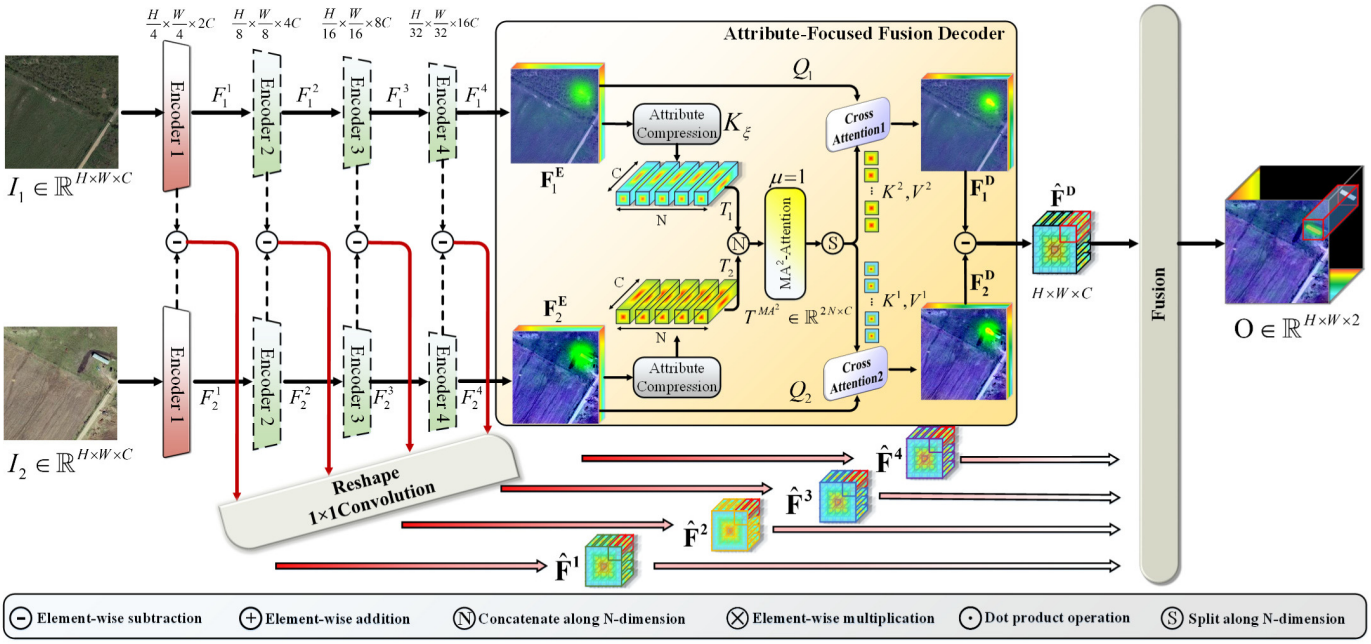



Fig. 2. Overview of the MNDAF-Net.

Transformer network that integrates temporal, spectral, and semantic information to improve the accuracy of attribute change understanding. Hassani and Shi [62] proposed a sparse global attention pattern that extends the attention range by increasing the step size (dilation) while maintaining the computational burden unchanged.

In summary, both CNNs and Transformers offer distinct advantages for attribute change understanding tasks. CNNs excel in local feature extraction and multiscale information fusion, while Transformers are particularly effective for modeling global dependencies, making them well-suited for attribute change understanding in complex scenes. However, existing methods still face challenges in adaptive neighborhood attribute capture and fine-grained feature fusion. To address these issues, we propose MNDAF-Net, leveraging the strengths of both approaches to tackle the complexities of high-resolution remote sensing attribute change understanding.

## III. Methodology

### A. Network Overview

MNDAF-Net employs a cascaded encoder–decoder structure to enable efficient understanding of scene attribute changes in bitemporal remote sensing images, as illustrated in Fig. 2. In the encoding phase, the network sequentially applies two feature modeling strategies: first, a convolutional local feature modeling module (Encoder 1) is used to extract foundational features; then, the neighborhood feature decoupling analysis (NFDA) modules (Encoders 2–4) capture neighborhood features across varying receptive fields, enhancing multiscale spatial information and fine-grained feature associations. These encoding modules effectively integrate local and global features from bitemporal images, providing a high-quality feature representation for subsequent processing.

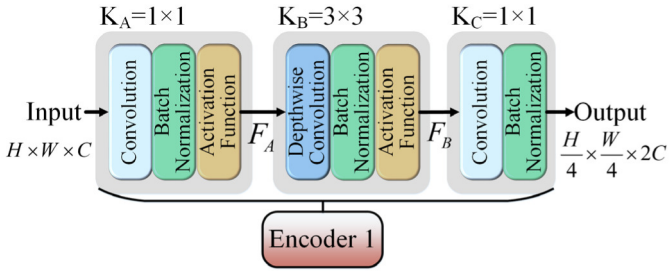



Fig. 3. Overview of the local feature modeling module.

Once encoding is complete, the extracted bitemporal features are fed into the AFF-Decoder designed for information interaction. During decoding, the network progressively restores spatial resolution while fusing feature differences between the bitemporal encoded representations at each level. This spatial fusion of feature differences enables the decoder to precisely focus on areas of change, enhancing both sensitivity and accuracy in change detection.

Finally, the multilevel fused features output from the AFF-Decoder are processed by a prediction head to generate a pixel-level scene attribute change map. This approach ensures that changes between bitemporal images are accurately captured and represented, providing robust support for change detection in remote sensing imagery.

### B. Encoder Module

*1) Local Feature Modeling:* In Encoder 1 (as shown in Fig. 3), feature extraction begins with a sequence of convolutional, batch normalization, and activation operations. A $1 \times 1$ convolution compresses the input feature map channels and integrates information across channels using a kernel $K_A$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

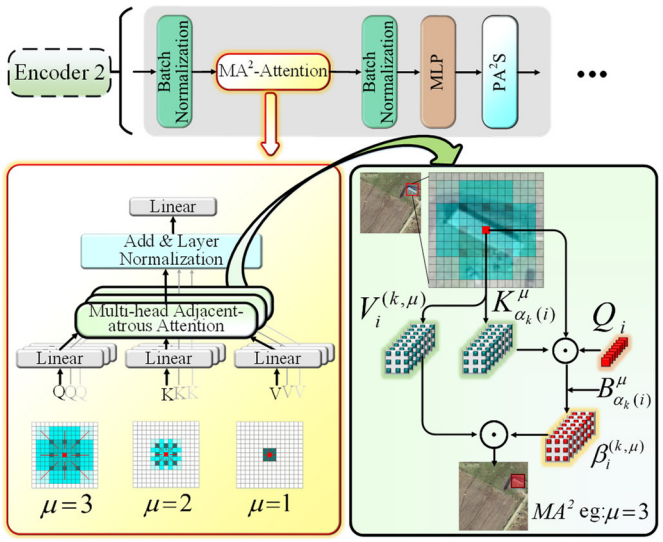



Fig. 4. Overview of the NFDA module.

resulting in a more enriched feature representation. The intermediate feature map $F_A$, processed with batch normalization and the GELU activation function, achieves standardized data distribution and enhanced nonlinear expressiveness.

To capture relationships within local image regions, Encoder 1 applies a $3 \times 3$ depthwise convolution to $F_A$, utilizing a kernel $K_B$ to focus on spatial details within each channel, preserving interchannel differences. The resulting feature map $F_B$, after additional batch normalization and activation, strengthens the representation of local region information. Finally, a subsequent $1 \times 1$ convolution further optimizes channel interactions, effectively merging local features to generate the output feature map $F_1$. This process ensures that Encoder 1 produces features with efficient compression, localized enhancement, and channel integration, establishing a robust foundation for subsequent encoding modules

$$\begin{aligned} F_A &= \mathrm{GELU}\left(\mathrm{BN}\left(I_A \times K_A\right)\right) \\ F_B &= \mathrm{GELU}\left(\mathrm{BN}\left(\mathrm{DepthwiseConv}\left(F_A, K_B\right)\right)\right) \\ F_1 &= \mathrm{BN}\left(F_B \times K_C\right). \end{aligned} \tag{1}$$

*2) NFDA Module:* Conventional CNN-based context encoders and multiscale fusion strategies rely on fixed convolutional receptive fields and simple feature concatenation, which often overlook the subtle contextual relationships between adjacent pixels. As a result, fine-grained semantic details may be lost during feature transformation. Meanwhile, standard self-attention mechanisms, such as those employed in Transformer architectures, effectively capture long-range global dependencies but tend to disregard immediate neighborhood cues, potentially weakening the representation of critical local variations. In contrast, Encoders 2–4 in our framework incorporate a cascaded NFDA module (as shown in Fig. 4), which leverages the $\mathrm{MA}^2$ module to constrain attention within local neighborhoods across multiple scales. This design preserves local structural context while dynamically expanding the receptive field through varying dilation rates.

The NFDA module introduces the $\mathrm{MA}^2$ mechanism, specifically designed to capture multiscale local and neighborhood-global relationships. The input to the $\mathrm{MA}^2$ layer is the feature map $F_1$, processed with batch normalization to ensure numerical stability and reduce instability during training. Following this, linear projections generate the query $Q$, key $K$, and value $V$ vectors, which model complex dependencies among features. $\mathrm{MA}^2$ applies various dilation rates $\mu$ (3, 2, and 1) to capture multiscale feature relationships, enhancing the model's receptive field and feature representation capabilities. Specifically, for a feature point $i$, its association with the $k$th nearest neighbor in its neighborhood is defined by the following formula:

$$\beta_i^{(k,\mu)} = Q_i \cdot \left(K_{\alpha(i)}^{k,\mu}\right)^{\mathrm{T}} + B_{\alpha(i)}^{k,\mu} \tag{2}$$

where $\alpha(i)$ represents the relationship between a feature point and its $k$th neighboring feature at dilation rate $\mu$, while $B_{\alpha(i)}^{k,\mu}$ denotes the relative positional deviation of the neighborhood, and it is a learnable parameter. This positional offset is used to better capture spatial relationships among features. By introducing various dilation rates, the $\mathrm{MA}^2$ layer dynamically adjusts its receptive field to accommodate different feature scales, enhancing the capture of both global and local information in the image

$$\mathrm{MA}^2 = \mathrm{softmax}\left(\frac{\beta_i^{k,\mu}}{\sqrt{d}}\right) \times V_{\alpha(i)} \tag{3}$$

where $d$ acts as a scaling factor to ensure numerical stability in calculations, while $V_{\alpha(i)}$ represents the neighborhood features of the $i$th position. The output of the $\mathrm{MA}^2$ layer undergoes layer normalization and residual connection, maintaining feature stability and continuity throughout the deep network. This structure enables the model to dynamically adapt to multiscale spatial dependencies, effectively capturing complex relationships among neighborhood features. As a result, it enhances the model's performance in interpreting intricate patterns within remote sensing imagery, addressing the demands of complex scene understanding tasks.

*3) $\mathrm{PA}^2\mathrm{S}$ Layer:* Unlike traditional multiscale context modules, the $\mathrm{PA}^2\mathrm{S}$ module (as illustrated in Fig. 5) aggregates global spatial information by combining large-kernel depthwise convolution with dilated convolution. It further incorporates a channel–spatial pooling strategy to generate adaptive attention weights over the feature map. This semantic compression mechanism selectively emphasizes salient features in change-sensitive regions while suppressing redundant information, effectively addressing the limitations of information loss and coarse feature merging observed in existing fusion approaches

$$\begin{aligned} F_{\mathrm{MA}^2}^{5\times5} &= \mathrm{conv}_{5\times5}\left(F_{\mathrm{MA}^2}\right) \\ F_{\mathrm{MA}^2}^{7\times7} &= \mathrm{conv}_{7\times7}^{\mu=3}\left(F_{\mathrm{MA}^2}^{5\times5}\right). \end{aligned} \tag{4}$$

The features are then passed through a fully connected layer for a nonlinear transformation, further enhancing

$$\begin{aligned} F_a^{5\times5} &= \mathrm{Dropout}\left(\mathrm{GELU}\left(\mathrm{BN}\left(F_{\mathrm{MA}^2}^{5\times5}\right)\right)\right) \\ F_b^{7\times7} &= \mathrm{Dropout}\left(\mathrm{GELU}\left(\mathrm{BN}\left(F_{\mathrm{MA}^2}^{7\times7}\right)\right)\right). \end{aligned} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

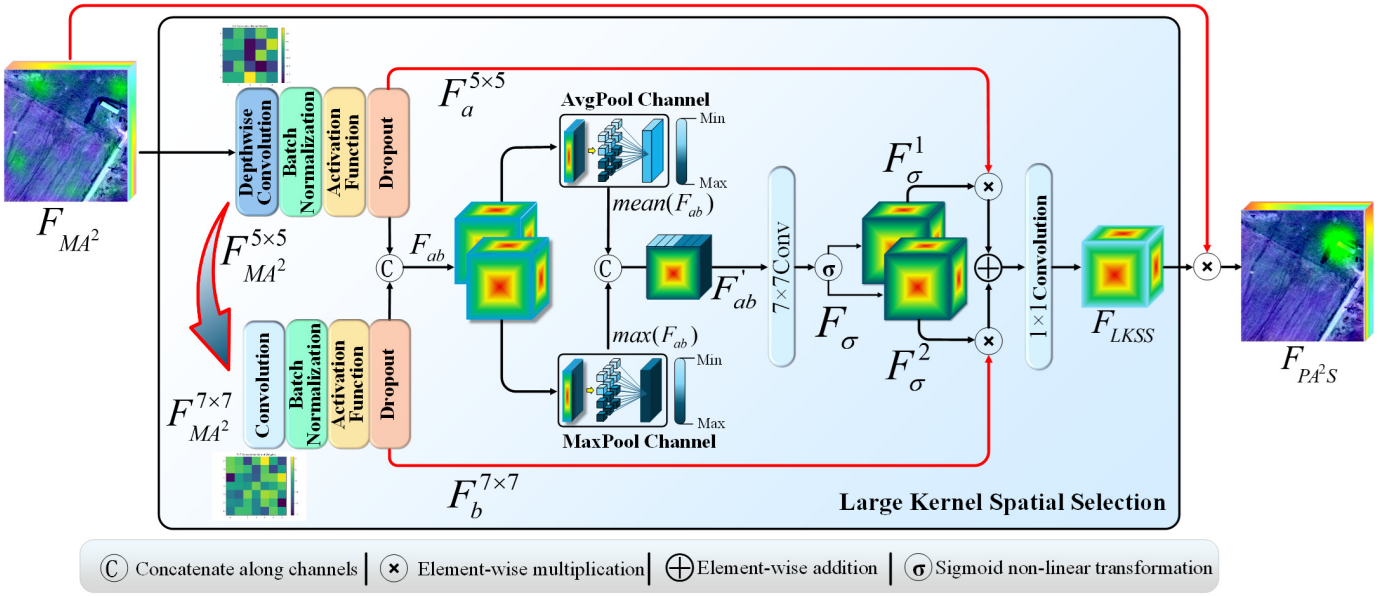



Fig. 5. Overview of the PA$^2$S module.

During the fusion of features along the channel and spatial dimensions, a concatenation operation is applied to create unified feature representations $F_{ab}$ and $F'_{ab}$. This representation is then processed with average pooling and max pooling to generate a selective weight matrix, which enables weighted fusion across spatial and channel dimensions

$$F_{ab} = \text{Concat}_{\text{channel}}\left(F_a^{5\times5}, F_b^{7\times7}\right) \tag{6}$$

$$F'_{ab} = \text{Concat}_{\text{channel}}\left(\text{mean}\left(F_{ab}\right), \text{max}\left(F_{ab}\right)\right). \tag{7}$$

Through these pooling operations, the PA$^2$S layer captures an overview of interchannel feature information and significance at a global scale, aiding the model in identifying the most representative features. This is followed by a $7 \times 7$ convolution layer that generates a selective weight matrix $F_\sigma$, which applies selective weighting to the features across both spatial and channel dimensions

$$F_\sigma = \text{sigmoid}\left(\text{conv}_{7\times7}\left(F'_{ab}\right)\right). \tag{8}$$

After normalization of $F_\sigma$ with the sigmoid function, the importance of each channel and spatial position is mapped to the range [0, 1], reflecting the relative significance of different features at both global and local scales. Subsequently, a $1 \times 1$ convolution is applied to the output feature to adjust its dimensions, ensuring that the output feature dimensions $F_{\text{LKSS}}$ are consistent with those of the input

$$F_{\text{LKSS}} = \text{conv}_{1\times1}\left(F_a^{5\times5} \otimes F_\sigma^1 + F_b^{7\times7} \otimes F_\sigma^2\right). \tag{9}$$

The final output of the PA$^2$S module is the elementwise product between the input feature $F_{\text{MA}^2}$ and $F_{\text{LKSS}}$, resembling a residual structure [63], [64]

$$F_{PA^2S} = F_{\text{MA}^2} \otimes F_{\text{LKSS}}. \tag{10}$$

### C. AFF-Decoder Module

Up to this point, the feature matrices following Encoder 4 are upsampled to the spatial resolution $H \times W \times C$ using deconvolution and $1 \times 1$ convolution operations, resulting in features $\mathbf{F_1^E}$ and $\mathbf{F_2^E}$.

In the AFF-Decoder stage, to effectively capture the temporal evolution of corresponding images, an attribute compression (AC) module is first employed to represent input image features as a compact set of high-level semantic attribute tokens. The core idea of this module is to compress the entire input feature map into a limited number of "visual attributes," thereby summarizing image content at the semantic level to enable more efficient subsequent processing and highlight key changing regions. After further excavating the compressed semantic attribute tokens through the MA $^2$ layer, they are distributed across different cross-attention mechanisms to achieve dual-temporal feature interaction learning. Ultimately, the interaction features undergo reconstruction and difference calculation to generate feature-level difference maps, providing high-discriminative semantic representations for change detection.

Specifically, pointwise convolution is applied to the input feature $\mathbf{F_i^E} \in \mathbb{R}^{HW\times C}$ ($i = 1$ and 2) for initial feature extraction. Then, the AC module uses pointwise convolution to learn $N$ semantic attribute groups from the input features, with each group representing an abstract semantic concept. Following this, a softmax function is applied across the spatial dimensions of each attribute group, producing an attribute correlation map $\xi_i \in \mathbb{R}^{HW\times N}$ that reflects the relationship between each location in the image and the corresponding semantic attribute group.

These attribute correlation maps are then used to compute a weighted average of the original feature map, resulting in a compact attribute set of size $N$, referred to as the attribute tokens $T_i \in \mathbb{R}^{N\times C}$. This AC process can be expressed as follows:

$$\begin{aligned} T_i &= (\xi_i)^{\text{T}}\, \mathbf{F_i^E} \\ &= \left(\text{softmax}\left(\boldsymbol{\nu}\left(F_i^E, K_\xi\right)\right)\right)^{\text{T}} \mathbf{F_i^E} \end{aligned} \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

where $\upsilon(\cdot)$ represents pointwise convolution with learnable weights $K_\xi \in \mathbb{R}^{C\times N}, N = 4$ (consistent with [21] and [22]). Ultimately, the AC module produces a set of compressed semantic attribute tokens $T_i$ that effectively represent high-level attribute information of the input image. Through this module, the model shifts its focus from pixel-level processing of all details to concentrating on a few key attributes, significantly reducing the complexity of subsequent computations. This compression approach not only preserves essential semantic information but also filters out unnecessary low-level details, providing high-quality input features for the subsequent $MA^2$ module to better capture the global semantic relationships between dual-temporal images.

Once the dual-temporal semantic attribute tokens $T_1$ and $T_2$ are generated through the AC module, the goal is to leverage $MA^2$ to fully explore global attribute relationships within the token-based temporal context, thereby creating richly contextualized attribute token representations for each temporal instance. This attribute extraction process, denoted as $T_{MA^2} \in \mathbb{R}^{2N\times C}$, is represented by the following equation:

$$T^{MA^2} = MA^2\left(N\left(T_1, T_2\right)\right) \tag{12}$$

where $N(\cdot)$ represents the concatenation of $T_1$ and $T_2$ along the $N$-dimension. Subsequently, $T^{MA^2}$ is split again along the $N$-dimension and used as the keys $(K)$ and values $(V)$ for cross attention. The definitions of the query, key, and value for the two parallel cross-attention mechanisms are as follows:

$$\begin{aligned}
&\text{cross-attention}_1 : Q_1 = \mathbf{F}_1^{\mathbf{E}}, \quad K_1 = T_2^{MA^2}, \quad V_1 = T_2^{MA^2} \\
&\text{cross-attention}_2 : Q_2 = \mathbf{F}_2^{\mathbf{E}}, \quad K_2 = T_1^{MA^2}, \quad V_2 = T_1^{MA^2}.
\end{aligned} \tag{13}$$

Finally, the two sets of interchanged feature maps are reshaped into 2-D feature maps $\mathbf{F}_1^{\mathbf{D}}, \mathbf{F}_2^{\mathbf{D}} \in \mathbb{R}^{H\times W\times C}$. These maps are then subtracted to produce a feature-level difference map $\hat{\mathbf{F}}^{\mathbf{D}} = |\mathbf{F}_1^{\mathbf{D}} - \mathbf{F}_2^{\mathbf{D}}|$. $\hat{\mathbf{F}}^{\mathbf{D}}$ serves as the decoder's final output and is utilized as one of the inputs for the prediction stage.

### D. Prediction Head

Leveraging the high-level semantic attribute features extracted by MNDAF-Net, we employ a simple convolutional structure to combine the decoder's output $\hat{\mathbf{F}}^{\mathbf{D}}$ with the difference maps ($\hat{\mathbf{F}}^1, \hat{\mathbf{F}}^2, \hat{\mathbf{F}}^3$, and $\hat{\mathbf{F}}^4$) at each encoder stage. This combined representation is then fed forward as input to the subsequent prediction process

$$\begin{aligned}
\hat{\mathbf{F}}^1 &= \left|F_1^1 - F_2^1\right| \\
\hat{\mathbf{F}}^2 &= \left|F_1^2 - F_2^2\right| \\
\hat{\mathbf{F}}^3 &= \left|F_1^3 - F_2^3\right| \\
\hat{\mathbf{F}}^4 &= \left|F_1^4 - F_2^4\right|.
\end{aligned} \tag{14}$$

Finally, the prediction head is used to generate the predicted attribute change probability $\mathbf{O} \in \mathbb{R}^{H\times W\times 2}$

$$\mathbf{O} = \text{softmax}\left(\text{conv}\left(\hat{\mathbf{F}}^1 \oplus \hat{\mathbf{F}}^2 \oplus \hat{\mathbf{F}}^3 \oplus \hat{\mathbf{F}}^4 \oplus \hat{\mathbf{F}}^{\mathbf{D}}\right)\right) \tag{15}$$

where $\oplus$ represents the elementwise addition, $\text{conv}(\cdot)$ represents the attribute change classifier, and $\text{softmax}(\cdot)$ is the softmax function applied pixelwise across the channel dimension of the classifier's output.

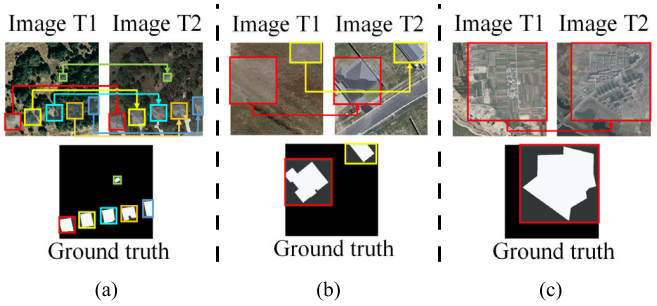



Fig. 6. Sample images from three datasets. (a) Sourced from the LEVIR-CD. (b) Sourced from the WHU-CD. (c) Sourced from the DSIFN-CD.

### E. Loss Function

In this article, the change detection task is formulated as a binary classification problem, where each pixel is categorized as either "changed" or "unchanged." We employ the classical binary cross-entropy loss to guide the training process of the model.

Specifically, the model outputs a 2-D change map, where the first dimension represents the probability or confidence of unchanged regions, and the second dimension represents the probability of changed regions. To compare the model output with the ground truth, the original single-channel label image is expanded to a two-channel format, where each pixel label is one-hot encoded as $\mathcal{G} \in \mathbb{R}^{H\times W\times 2}$. $H$ and $W$ denote the height and width of the input image, respectively. The first channel of the one-hot encoding corresponds to the unchanged category, while the second channel corresponds to the changed category. This encoding scheme ensures that differences between categories can be accurately captured during loss computation.

The loss function is calculated by comparing the model output with the one-hot encoded ground truth, formulated as

$$\mathcal{L}_{\text{BCE}}\left(\mathcal{G}, \mathcal{P}\right) = -\frac{1}{H\times W}\sum_{i=1}^{H\times W}\left(\mathcal{G}\left(i\right)\log \mathcal{P}\left(i\right)\right) \tag{16}$$

where $\mathcal{G}$ denotes the one-hot encoded ground truth and $\mathcal{P}$ denotes the model's predicted output. For each pixel $i$, the loss is computed by performing a weighted summation of the log probabilities from the model output.

## IV. Data Preparation and Experimental Preparation

### A. Materials and Data Preparation

To validate the effectiveness of the proposed method, we selected three challenging datasets: LEVIR-CD [65], WHU-CD [66], and DSIFN-CD [67]. Fig. 6 illustrates attribute change detection examples from each dataset.

*1) LEVIR-CD:* The LEVIR-CD dataset is specifically designed for building change detection tasks in remote sensing imagery, aimed at advancing technology in this area. This dataset includes 637 pairs of remote sensing images captured at two different times (T1 and T2), sourced from Google Earth, with a spatial resolution of 0.5 m. Each image pair is sized $1024 \times 1024$ pixels and covers 31 333 instances of building changes, providing substantial support for building change

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

detection tasks. The dataset is split into a training set (445 image pairs), a validation set (64 image pairs), and a testing set (128 image pairs), establishing a comprehensive foundation for model training, parameter optimization, and performance evaluation.

*2) WHU-CD:* The WHU-CD dataset is primarily used for change detection tasks, containing aerial images from different time periods and covering an area of approximately 20.5 $km^2$ with 12 796 buildings. The original images have a spatial resolution of 0.075 m/pixel, with a full size of $32\,507 \times 15\,354$ pixels. To facilitate model training, the original images were divided into nonoverlapping image blocks of $256 \times 256$ pixels. The dataset is partitioned into 5947 training image blocks, 743 validation blocks, and 744 testing blocks.

*3) DSIFN-CD:* The DSIFN-CD dataset focuses on urban change detection, encompassing remote sensing images from six major cities in China (Chengdu, Beijing, Shenzhen, Chongqing, Wuhan, and Xi'an). This dataset documents changes across various land cover types, including buildings, roads, and farmland. Each original image pair is $512 \times 512$ pixels and was further divided into nonoverlapping image blocks of $256 \times 256$ pixels. The dataset consists of 14 400 training image pairs, 1360 validation pairs, and 192 testing pairs.

The selection of these datasets enables a robust evaluation of the proposed method across diverse geographic regions and change types, demonstrating the method's robustness and applicability across a range of conditions.

### B. Evaluation Metrics

Following previous research methodologies, we adopted the five most commonly used metrics to evaluate detection performance: $F1$-score (F1), intersection over union (IoU), overall accuracy (OA), precision (Pre), and recall (Rec). These metrics are defined as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad F1 = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre}+\text{Rec}}$$
$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}}, \quad \text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FN}+\text{TN}+\text{FP}}. \quad (17)$$

### C. Implementation Details

During model training, we utilized a single NVIDIA A100 GPU with a batch size of 64. The network optimization was conducted using the AdamW optimizer with a weight decay ($L2$ regularization) coefficient of $1E^{-2}$. An initial learning rate of $1E^{-4}$ was set and dynamically adjusted using a cosine annealing scheduler.

Input images were preprocessed and resized to $H \times W \times C$ while maintaining the original three-channel structure. Each convolutional layer employed a stride of 2, reducing the spatial dimensions of feature maps by half and doubling the channel count. The output dimensions for Encoders 2–4 were $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$, with corresponding channel counts of $2C$, $4C$, and $8C$, respectively. Notably, in the dual-branch encoding layers, each layer shared the same parameters.

TABLE I

COMPARISON OF EXPERIMENTAL RESULTS (%) WITH STATE-OF-THE-ART METHODS ON THE LEVIR-CD DATASET. BOLD MEANS THE BEST PERFORMANCE. UNDERLINE REPRESENTS THE SECOND-BEST RESULT

| Method | Year | Pre | Rec | F1 | IoU | OA |
|---|---|---|---|---|---|---|
| IFNet[67] | 2020 | 94.02 | 82.93 | 88.13 | 78.77 | 98.87 |
| STANet[65] | 2020 | 83.81 | 91.00 | 87.26 | 77.40 | 98.66 |
| BIT[21] | 2022 | 89.18 | 87.17 | 88.16 | 78.83 | 98.82 |
| SNUNet-CD[68] | 2022 | 89.24 | 89.37 | 89.31 | 80.68 | 98.92 |
| ChangeFormer[51] | 2022 | 92.05 | 88.80 | 90.40 | 82.48 | 99.04 |
| WNet[53] | 2023 | 91.99 | 88.51 | 90.22 | 82.18 | 99.02 |
| VcT[22] | 2024 | 91.16 | 90.18 | 90.67 | 82.93 | 99.06 |
| ChangeCLIP[69] | 2024 | 90.82 | 89.77 | 90.29 | 82.30 | 99.02 |
| BASNet[70] | 2024 | 92.66 | 88.81 | 90.69 | 82.96 | 99.07 |
| ASGCN[41] | 2024 | 89.34 | 92.50 | 90.83 | 83.21 | 99.08 |
| AMFNet[71] | 2024 | **94.77** | 91.15 | 90.79 | 83.13 | 99.07 |
| CGMMA[42] | 2024 | 93.68 | 89.04 | 91.30 | 83.99 | 99.14 |
| GCFormer[55] | 2024 | 92.94 | 90.31 | 91.60 | 84.51 | 99.16 |
| HCGNet[44] | 2024 | 90.82 | **92.42** | 91.61 | 84.52 | 99.14 |
| RFANet[72] | 2024 | 92.27 | 91.34 | <u>91.81</u> | 84.85 | <u>99.17</u> |
| MNDAF-Net (our) | 2025 | <u>94.27</u> | <u>91.93</u> | **93.09** | **85.42** | **99.19** |

The decoder's input and output dimensions matched the original input size, set at $H \times W \times C$. In the decoder layers, the $MA^2$ module with a dilation rate of $\mu = 1$ was employed to effectively capture multiscale feature information. The kernel size, stride, and padding of the pointwise convolution are all set to 1.

## V. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Contrast Evaluation for the State-of-the-Art Methods

We conducted a comprehensive comparison of our proposed method with several state-of-the-art change detection models for remote sensing imagery developed in recent years. These models include advanced techniques that combine Transformer and CNN architectures, such as IFNet [67], STANet [65], BIT [21], SNUNet-CD [68], ChangeFormer [51], WNet [53], and VcT [22]. In addition, we evaluated methods leveraging graph convolution and attention mechanisms, including ASGCN [41], CGMMA [42], and GCFormer [55], which have shown effectiveness in modeling global context and structural information in complex remote sensing scenarios.

Furthermore, we considered recent models incorporating adaptive attention modules and hybrid architectures, such as HCGNet [44] and RFANet [69]. These methods represent the latest progress in integrating multiscale feature extraction with adaptive attention to capture intricate spatial changes. By integrating multiscale information and focusing on both local and global dependencies, these models effectively capture complex attribute changes, providing robust support for improving the accuracy and robustness of change detection in remote sensing imagery.

In the comparative experiments, MNDAF-Net demonstrated significant performance improvements across multiple key

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

TABLE II

COMPARISON OF EXPERIMENTAL RESULTS (%) WITH STATE-OF-THE-ART METHODS ON THE WHU-CD DATASET. BOLD MEANS THE BEST PERFORMANCE. UNDERLINE REPRESENTS THE SECOND-BEST RESULT

| Method | Year | Pre | Rec | F1 | IoU | OA |
|---|---|---|---|---|---|---|
| IFNet[67] | 2020 | 78.00 | 70.81 | 74.23 | 59.03 | 92.53 |
| STANet[65] | 2020 | 79.37 | 85.50 | 82.32 | 69.95 | 98.52 |
| SNUNet-CD[68] | 2022 | 85.60 | 81.49 | 83.50 | 71.67 | 98.71 |
| BIT[21] | 2022 | 86.64 | 81.48 | 83.98 | 72.39 | 98.75 |
| ChangeFormer[51] | 2022 | 91.83 | 88.02 | 89.88 | 81.63 | 99.12 |
| VcT[22] | 2024 | 89.39 | 89.77 | 89.58 | 81.12 | 99.18 |
| WNet[53] | 2023 | 92.37 | 90.15 | 91.25 | 83.91 | 99.31 |
| GCFormer[55] | 2024 | 91.81 | 90.15 | 90.97 | 83.43 | 99.29 |
| RFANet[72] | 2024 | 92.97 | 89.58 | 91.24 | 83.90 | 99.33 |
| ASGCN[41] | 2024 | 92.42 | 90.14 | 91.26 | 83.93 | 99.32 |
| CGMMA[42] | 2024 | 95.25 | 90.48 | 92.81 | 86.58 | 99.32 |
| HCGNet[44] | 2024 | **98.84** | 91.64 | 93.69 | 88.14 | 99.40 |
| ChangeCLIP[69] | 2024 | 96.02 | **93.58** | 94.78 | 90.08 | **99.52** |
| MNDAF-Net (our) | 2025 | 97.73 | 92.56 | **95.07** | **90.24** | 99.38 |

TABLE III

COMPARISON OF EXPERIMENTAL RESULTS (%) WITH STATE-OF-THE-ART METHODS ON THE DSIFN-CD DATASET. BOLD MEANS THE BEST PERFORMANCE. UNDERLINE REPRESENTS THE SECOND-BEST RESULT

| Method | Year | Pre | Rec | F1(%) | IoU | OA |
|---|---|---|---|---|---|---|
| IFNet[67] | 2020 | 67.86 | 53.94 | 60.10 | 42.96 | 87.83 |
| STANet[65] | 2020 | 67.71 | 61.68 | 64.56 | 47.66 | 88.49 |
| SNUNet-CD[68] | 2022 | 60.60 | 72.89 | 66.18 | 49.45 | 87.34 |
| BIT[21] | 2022 | 68.36 | 70.18 | 69.26 | 52.97 | 89.41 |
| ChangeFormer[51] | 2022 | 88.48 | 84.94 | 86.67 | 76.48 | 95.59 |
| VcT[22] | 2024 | 83.91 | 66.47 | 74.18 | 58.95 | 92.14 |
| RFANet[72] | 2024 | 84.90 | 61.13 | 71.08 | 55.14 | 91.55 |
| ASGCN[41] | 2024 | 87.73 | 88.33 | 88.02 | 78.61 | **95.92** |
| GCFormer[55] | 2024 | 87.31 | 88.02 | 87.66 | 78.04 | 95.79 |
| WNet[53] | 2023 | 89.13 | **91.51** | **90.81** | 78.95 | 95.03 |
| MNDAF-Net (our) | 2025 | **91.92** | 89.36 | 90.62 | **80.13** | 95.81 |

metrics. Tables I–III present the experimental results on the LEVIR-CD, WHU-CD, and DSIFN-CD public datasets, respectively. Compared with the current state-of-the-art models, MNDAF-Net consistently outperformed on these datasets, highlighting its efficacy, adaptability, and robustness across varied remote sensing scenarios.

*1) LEVIR-CD:* On the LEVIR-CD dataset, MNDAF-Net achieved an F1-score of 93.09%, an IoU of 85.42%, and an OA of 99.19%. These values mark an improvement of 1.28%, 0.57%, and 0.02%, respectively, over the prior top-performing RFANet model [72]. This notable performance increase is attributed to MNDAF-Net's innovative design in the NFDA module, which emphasizes multiscale feature extraction and AC paired with cross learning during the decoding phase. The $MA^2$ module's multihead adjacent dilation attention mechanism enhances the model's capacity for capturing both local details and global features, while the $PA^2S$ module further boosts focus on critical change regions, significantly improving change detection accuracy on the LEVIR-CD dataset.

*2) WHU-CD:* On the WHU-CD dataset, MNDAF-Net achieved an F1 of 95.07%, an IoU of 90.24%, and an OA of 99.38%, outperforming all competing methods and demonstrating outstanding results, particularly in the F1 and IoU metrics. This outcome indicates that MNDAF-Net's multiscale neighborhood feature decoupling and adaptive attribute sensing designs achieve robust adaptability across varying scene complexities. The complexity of the WHU-CD dataset requires models to excel at interpreting highly variable scenes and high-resolution features, and MNDAF-Net, with its meticulously designed modules, exhibited clear advantages on this dataset.

*3) DSIFN-CD:* On the more challenging DSIFN-CD dataset, MNDAF-Net also performed well, achieving an F1-score of 90.62%, an IoU of 80.13%, and an OA of 95.81%. Although MNDAF-Net's F1-score and OA score were slightly below those of the best-performing WNet [53], it exceeded WNet by 1.18% in IoU, underscoring its adaptability and robustness in complex remote sensing scenarios. The DSIFN-CD dataset presents intricate scene changes and subtle distinctions, yet MNDAF-Net consistently maintained high interpretation accuracy, enabled by its effective learning of neighborhood dependencies and comprehensive multiscale contextual modeling.

To provide a more intuitive comparison of model performance across the three benchmark datasets, we present visualized results in Figs. 7–9. These comparisons offer a clearer understanding of the strengths and weaknesses of each method. To further investigate the effectiveness of MNDAC-Net, we analyze the contribution of its core architectural components to the overall performance.

The $MA^2$ module introduces a multihead neighborhood-aware dilated attention mechanism that constrains attention within multiscale local regions. This design preserves fine-grained spatial details while simultaneously establishing sparse long-range dependencies across structural boundaries. As a result, the model's sensitivity to localized changes is significantly enhanced. In parallel, the $PA^2S$ module performs channel–spatial joint modeling to compress high-level semantic features during decoding. By emphasizing salient regions, this module refines the model's focus on meaningful change areas. Together, $MA^2$ and $PA^2S$ operate in complementary stages of the network encoding and decoding, forming a cohesive context modeling framework that balances local precision with global consistency. Additional ablation studies in Section V-B provide detailed quantitative evidence of their effectiveness. Notably, despite the integration of these two function-specific modules, the overall network maintains a manageable level of architectural complexity. The parameter count and computational load remain substantially lower than those of Transformer-based architectures relying solely on global attention mechanisms. Experimental results across the LEVIR-CD, WHU-CD, and DSIFN-CD datasets demonstrate that MNDAC-Net consistently achieves leading performance. More importantly, its generalization and robustness extend beyond ideal data conditions, making it well-suited for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

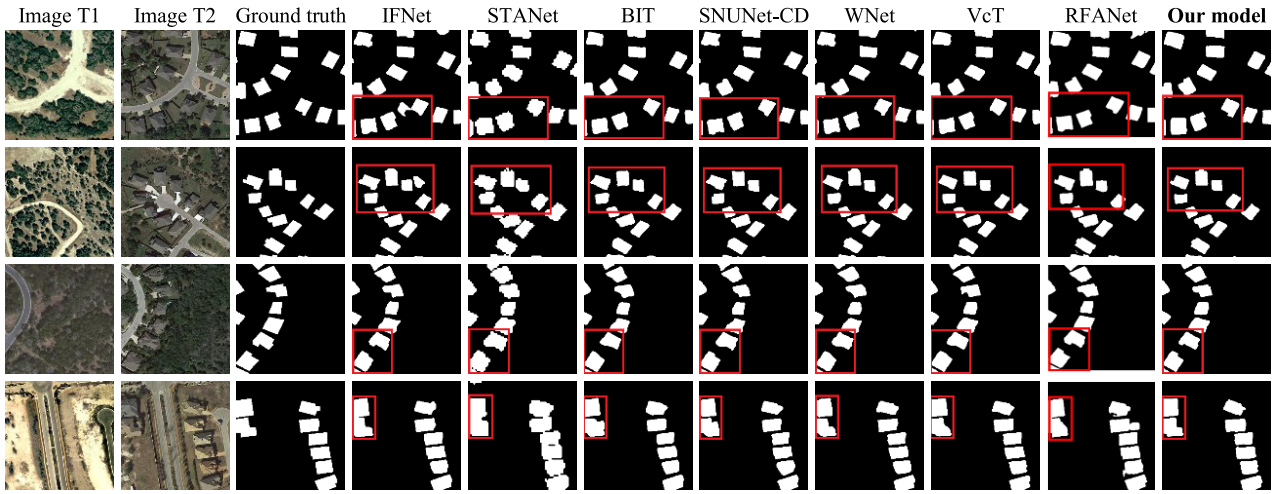



Fig. 7. Visual comparison of the experimental results on the LEVIR-CD dataset.

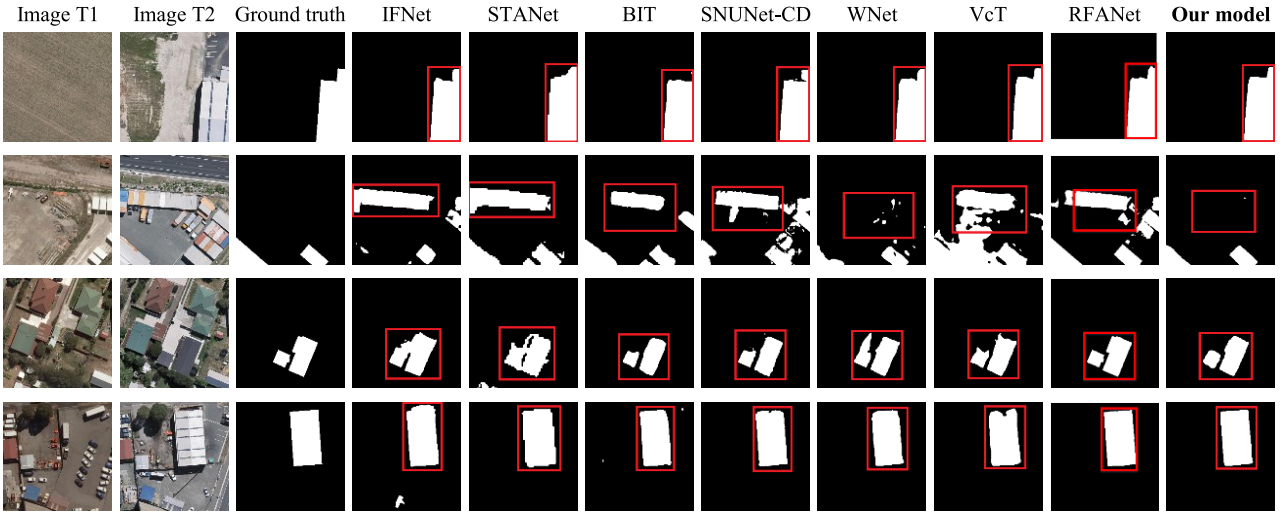



Fig. 8. Visual comparison of the experimental results on the WHU-CD dataset.

scenarios involving domain shifts, limited training samples, or complex change categories.

### B. Ablation Study

*1) Hyperparametric Ablation Experiment:* To enhance the interpretability of the model, Fig. 10 provides visual analysis results from both the attention mechanism and the module ablation perspective. The top rows present input bitemporal images (T1 and T2), ground-truth change masks, and corresponding heatmaps produced by the $MA^2$ module with different dilation rates ($\mu = 3$ and $\mu = 2$), as well as the final attention distribution of the full MNDAF-Net. These visualizations reveal how the attention mechanism selectively focuses on key change regions: larger dilation ($\mu = 3$) captures broader contextual semantics, while smaller dilation ($\mu = 2$) emphasizes fine-grained local variations. The attention map of MNDAF-Net demonstrates a balanced integration of both global and local cues, highlighting the change region with high confidence.

TABLE IV
ABLATION STUDY OF $\mu$ ON LEVIR-CD DATASETS. BOLD MEANS THE BEST PERFORMANCE

| $\mu$ | | | F1 | IoU | OA |
|---|---|---|---|---|---|
| Encder2 | Encder3 | Encder4 | | | |
| 3 | 3 | 3 | 89.85 | 81.66 | 98.97 |
| 2 | 2 | 2 | 90.43 | 82.53 | 99.04 |
| 1 | 1 | 1 | 91.15 | 83.66 | 99.11 |
| 3 | 2 | 1 | **93.09** | **85.42** | **99.19** |

The bottom section compares the predicted change maps under five different configurations: $MA^2$ with $\mu = 3$, $MA^2$ with $\mu = 2$, the full model without $MA^2$, the full model without $PA^2S$, and the complete MNDAF-Net. The comparison clearly shows that removing either $MA^2$ or $PA^2S$ results in reduced accuracy and incomplete change boundaries, while the complete model yields the most precise and compact predictions. These results intuitively explain the role of each module and demonstrate their complementary effect in guiding the model to focus on meaningful change regions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

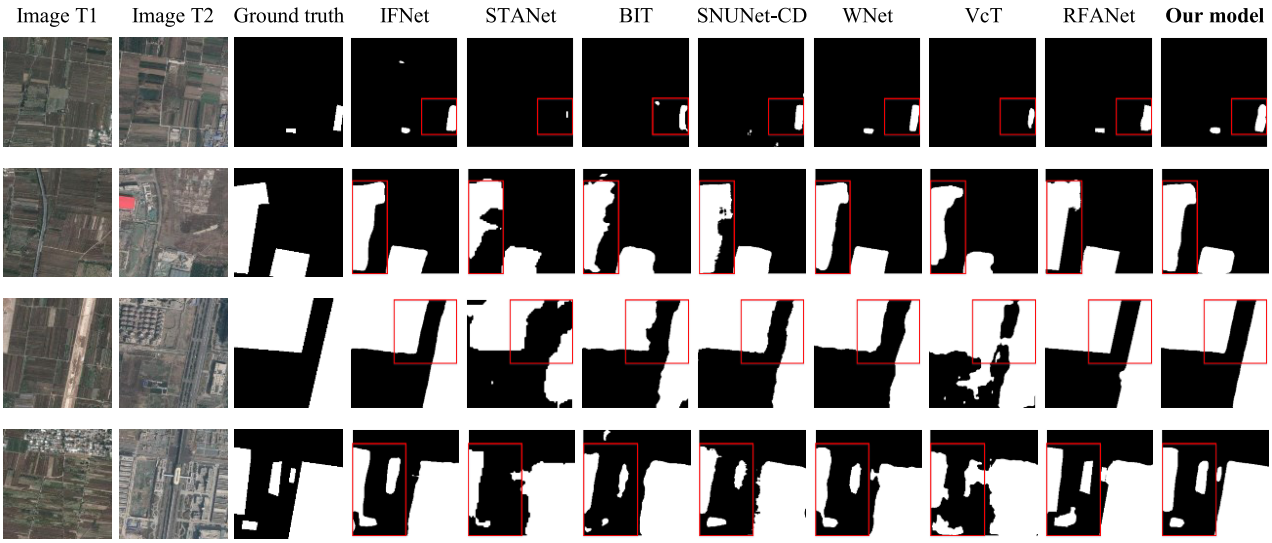



Fig. 9. Visual comparison of the experimental results on the DSIFN-CD dataset.

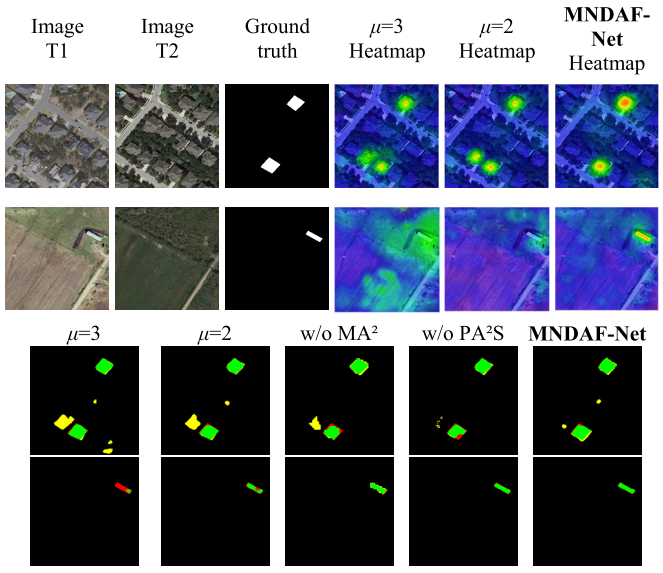



Fig. 10. Visualization of ablation study on LEVIR-CD datasets. True positives (TPs), FPs, false negatives (FNs), and true negatives (TNs) are represented in green, yellow, red, and black.

As shown in Table IV, the results vary across three fixed dilation rates (set at 3, 2, or 1). When $\mu = 1$, the MA$^2$ mechanism operates similar to traditional multihead self-attention, yielding an F1-score of 91.15%, IoU of 83.66%, and OA of 99.11%. Although this configuration performs well, it lacks the ability to capture multiscale features.

The results indicate that a progressively decreasing dilation rate configuration (where Encoder 2 uses $\mu = 3$, Encoder 3 uses $\mu = 2$, and Encoder 4 uses $\mu = 1$) achieves optimal performance, with an F1-score of 93.09%, IoU of 85.42%, and OA of 99.19%. This finding suggests that using a larger dilation rate in the early encoding stages helps the model capture richer global information, while decreasing the dilation rate in later stages focuses on fine-grained local features, thereby enhancing the model's understanding of change regions. This approach mirrors the logic of standard downsampling.

TABLE V

ABLATION STUDY OF PA$^2$S ON LEVIR-CD DATASETS. BOLD MEANS THE BEST PERFORMANCE

| PA²S | | F1 | IoU | OA |
|---|---|---|---|---|
| Depthwise convolution | Dilated Convolution | | | |
| 3×3 | 5×5 | 92.31 | 84.03 | 98.82 |
| 5×5 | 5×5 | 92.79 | 84.81 | 98.92 |
| 5×5 | 7×7 | **93.09** | **85.42** | **99.19** |
| 7×7 | 9×9 | 92.60 | 83.74 | 98.85 |

The innovation of the MA$^2$ mechanism lies in its multiscale feature modeling capability. Unlike standard multihead self-attention, MA$^2$ leverages various dilation rates to disentangle and model spatial relationships across different scales. This ablation experiment demonstrates that the multiscale feature extraction significantly enhances the detection of complex scene changes, validating the importance of multiscale characteristics in change detection tasks for bitemporal remote sensing images.

Table V presents the ablation study of the PA$^2$S module on the LEVIR-CD dataset. The experiment aims to evaluate the impact of different combinations of depthwise convolutions and dilated convolutions on model performance. As shown in the table, when employing a combination of a $5 \times 5$ depthwise convolution and a $7 \times 7$ dilated convolution, the model achieves optimal performance across all metrics, with an F1-score of 93.09%, an IoU of 85.42%, and an OA of 99.19%. This indicates that larger convolution kernels can more effectively capture global spatial information in remote sensing images while enhancing sensitivity to change regions.

In contrast, although the combination of $7 \times 7$ and $9 \times 9$ convolutions offers a larger receptive field, its performance in terms of F1-score and IoU shows a noticeable decline. This degradation is attributed to information loss or excessive smoothing caused by excessively large kernels during feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

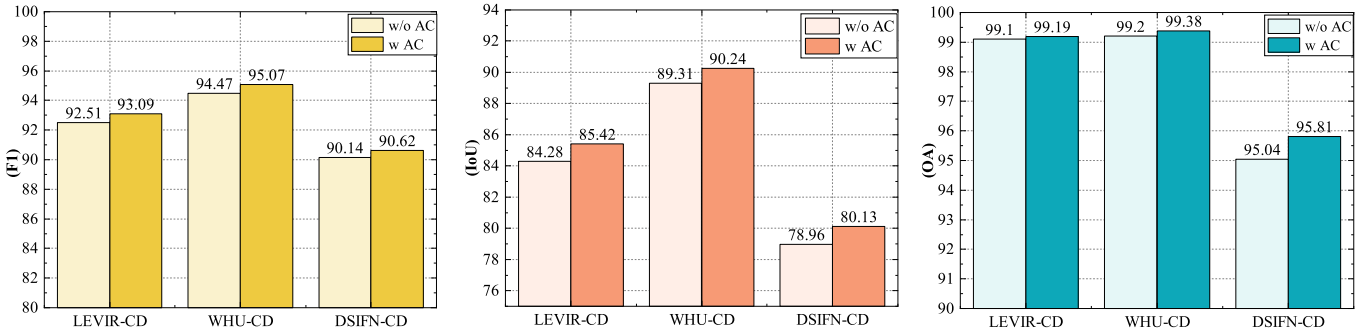



Fig. 11. Analysis results for AFF-Decoder structures.

TABLE VI
ABLATION STUDY RESULTS (%) WITH NFDA MODULE ON LEVIR-CD DATASETS. BOLD MEANS THE BEST PERFORMANCE

| MA² | PA²S | AFF Decoder | | F1 | IoU | OA |
|---|---|---|---|---|---|---|
| | | AC | Cro-A | | | |
| × | × | ✓ | ✓ | 88.06 | 78.64 | 98.83 |
| ✓ | × | ✓ | ✓ | 91.46 | 84.17 | 99.14 |
| × | ✓ | ✓ | ✓ | 91.15 | 83.66 | 99.11 |
| ✓ | ✓ | ✓ | ✓ | **93.09** | **85.42** | **99.19** |

TABLE VII
ABLATION STUDY RESULTS (%) WITH NFDA MODULE ON WHU-CD DATASETS. BOLD MEANS THE BEST PERFORMANCE

| MA² | PA²S | AFF Decoder | | F1 | IoU | OA |
|---|---|---|---|---|---|---|
| | | AC | Cro-A | | | |
| × | × | ✓ | ✓ | 91.27 | 84.15 | 99.13 |
| ✓ | × | ✓ | ✓ | 93.53 | 88.98 | 99.32 |
| × | ✓ | ✓ | ✓ | 92.98 | 88.42 | 99.26 |
| ✓ | ✓ | ✓ | ✓ | **95.07** | **90.24** | **99.38** |

extraction. Furthermore, smaller kernel combinations (e.g., $3 \times 3$ and $5 \times 5$) exhibit inferior performance compared with larger kernels, suggesting that small-scale convolutions are insufficient in capturing long-range contextual information and constructing a comprehensive global receptive field.

*2) Ablation Study of AFF-Decoder:* Fig. 11 presents the ablation results of the AC module across different datasets, including LEVIR-CD, WHU-CD, and DSIFN-CD. The results show that introducing the AC module consistently improves performance in terms of F1-score, IoU, and OA, regardless of the dataset. For instance, on the LEVIR-CD dataset, the F1-score and IoU increase by approximately 0.6% and 1.14%, respectively, when the AC module is used, along with a noticeable improvement in OA. Similar performance gains are observed on the WHU-CD and DSIFN-CD datasets, further validating the effectiveness of the AC module in enhancing semantic representation and facilitating global feature interaction.

These results demonstrate that the AC module not only effectively extracts and compresses high-level semantic information from input images, reducing the influence of redundant low-level details, but also significantly enhances the model's discriminative capability for change detection by optimizing the subsequent cross-temporal feature interaction.

TABLE VIII
ABLATION STUDY RESULTS (%) WITH NFDA MODULE ON DSIFN-CD DATASETS. BOLD MEANS THE BEST PERFORMANCE

| MA² | PA²S | AFF Decoder | | F1 | IoU | OA |
|---|---|---|---|---|---|---|
| | | AC | Cro-A | | | |
| × | × | ✓ | ✓ | 81.31 | 71.96 | 92.66 |
| ✓ | × | ✓ | ✓ | 88.58 | 78.52 | 94.49 |
| × | ✓ | ✓ | ✓ | 86.96 | 77.97 | 93.54 |
| ✓ | ✓ | ✓ | ✓ | **90.62** | **80.13** | **95.81** |

*3) Ablation Study of NFDA Module:* To validate the specific contributions of the $MA^2$ and $PA^2S$ modules within the NFDA module, ablation experiments were conducted on the LEVIR-CD, WHU-CD, and DSIFN-CD datasets, with results summarized in Tables VI–VIII, respectively. Notably, in the absence of the $MA^2$ module, traditional multihead self-attention was employed as a substitute to maintain structural integrity.

The results presented in Table VI for the LEVIR-CD dataset indicate that when both the $MA^2$ and $PA^2S$ modules are disabled, the model, using a standard ResNet18 encoder as in previous works [21], [22], [53], achieves an F1-score of 88.06% and an IoU of 78.64%. In contrast, when both modules are enabled, the model reaches its highest performance, with an F1-score of 93.09%, IoU of 85.42%, and OA of 99.19%. Removing the $PA^2S$ module while keeping $MA^2$ leads to a performance drop, with F1 and IoU decreasing to 91.46% and 84.17%, respectively. This result highlights the role of $PA^2S$ in refining feature selection and focusing on critical change regions. Similarly, removing the $MA^2$ module further reduces the F1 to 91.15% and IoU to 83.66%, underscoring its importance in capturing multiscale spatial information and handling complex scene changes.

Experimental results on the WHU-CD dataset (see Table VII) further support this trend. The baseline model achieves an F1-score of 91.27% and an IoU of 84.15%. Adding only the $PA^2S$ module improves the F1 to 93.53% and IoU to 88.98%. With only the $MA^2$ module, the F1 reaches 92.98% and IoU 88.42%. When both modules are used together, the model achieves the best performance, with an F1-score of 95.07% and an IoU of 90.24%. These results demonstrate that the combined local and global semantic modeling provided by $MA^2$ and $PA^2S$ enhances the model's adaptability and structural representation in complex, high-resolution scenes with diverse land cover types.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

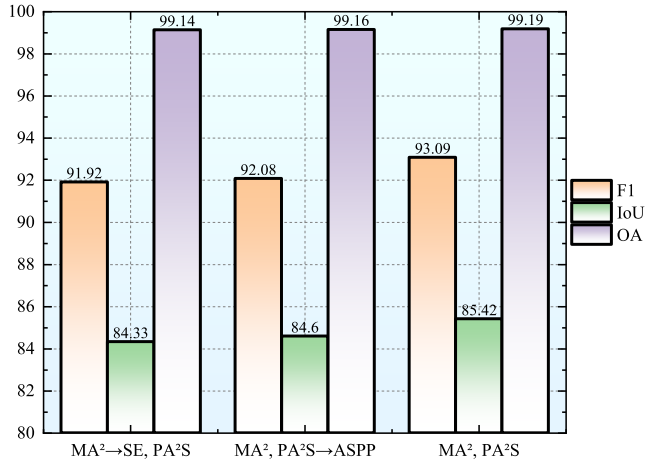



Fig. 12. Ablation study of the NFDA on LEVIR-CD datasets.

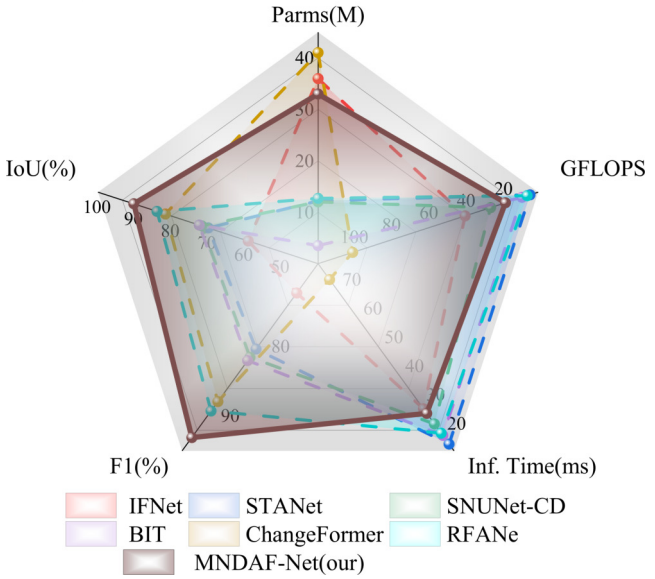



Fig. 13. Complexity comparison of different methods in terms of the Params, GFLOPs, inference time, $F1$, and IoU on WHU-CD datasets.

On the more challenging DSIFN-CD dataset (see Table VIII), the performance improvement is even more evident. The baseline model yields an F1-score of 81.31% and an IoU of 71.96%. Incorporating only the PA$^2$S module increases the F1 to 86.96% and IoU to 77.97%. Introducing only the MA$^2$ module results in an F1 of 88.58% and an IoU of 78.52%. When both modules are enabled, the model achieves an F1 of 90.62% and an IoU of 80.13%, clearly outperforming all other configurations. These findings confirm that in complex remote sensing scenarios with intricate change patterns and multiscale spatial structures, the joint application of MA$^2$ and PA$^2$S is especially beneficial.

As shown in Fig. 12, replacing either the MA$^2$ or PA$^2$S module with alternative components results in a noticeable decline in performance compared with the baseline model that integrates both modules. Specifically, substituting MA$^2$ with the SE module leads to a decrease of 1.17% in F1-score, 1.09% in IoU, and 0.05% in OA. Similarly, replacing PA$^2$S with the ASPP module results in a 1.01% drop in F1, a 0.82% reduction in IoU, and a 0.03% decrease in OA. The MA$^2$ module enhances the receptive field through a multihead adjacent dilated attention mechanism, allowing the model to simultaneously capture both local and global multiscale context. This design strengthens the model's ability to detect subtle boundary changes and structural details. However, when replaced by the SE module, which focuses only on channel-wise recalibration, the model loses its capability to model spatial neighborhood relationships. As a result, its sensitivity to fine-grained variations is reduced, leading to overall performance degradation. Similarly, the PA$^2$S module aggregates global spatial features using a combination of large-kernel and dilated convolutions, and it employs a channel–spatial pooling mechanism to generate adaptive attention weights. This helps the model emphasize salient characteristics in regions with significant change. In contrast, while the ASPP module also uses multiscale convolutional branches to increase the receptive field, it lacks an adaptive weighting mechanism. This limits its ability to enhance key change areas effectively, thereby constraining detection accuracy.

TABLE IX
MODELS' NUMBER OF PARAMS AND GFLOPS

| Method | Params(M) | GFLOPs | F1(%) | IoU(%) |
|---|---|---|---|---|
| BIT [21] | 3.50 | 10.63 | 83.98 | 72.39 |
| VcT [22] | 3.57 | 10.64 | 89.58 | 81.12 |
| SNUNet-CD [68] | 12.03 | 27.44 | 83.50 | 71.67 |
| STANet[65] | 12.21 | 6.35 | 82.32 | 69.95 |
| RFANet(Res18) [72] | 12.70 | 7.91 | 91.24 | 83.90 |
| CGMMA[42] | 17.19 | 41.78 | 92.81 | 86.58 |
| GCFormer [55] | 35.43 | 11.30 | 90.97 | 83.43 |
| HCGNet [44] | 38.68 | 14.18 | 93.69 | 88.14 |
| WNet [53] | 43.07 | 19.20 | 91.25 | 83.91 |
| IFNet [67] | 35.99 | 41.18 | 74.23 | 59.03 |
| ChangeFormer[51] | 41.03 | 101.43 | 89.88 | 81.63 |
| MNDAF-Net (our) | 32.98 | 19.52 | 95.07 | 90.24 |

In conclusion, the MA$^2$ module enhances the model's sensitivity to spatial structure variations and long-range dependencies through a multihead neighborhood-dilated attention mechanism. The PA$^2$S module complements this by applying semantic compression and joint channel–spatial enhancement, which improves focus on critical change regions while suppressing irrelevant background information. The integration of both modules consistently improves performance across datasets, demonstrating improved generalization and robustness in diverse change detection settings.

*4) Parameter Quantity Experiment:* Table IX presents the parameter count (Params) and computational complexity (GFLOPs) of various models, revealing each model's computational overhead and resource requirements in change detection tasks. As shown in Table IX and Fig. 13 for the WHU-CD benchmark, MNDAF-Net achieves the best balance of accuracy and efficiency among the surveyed change-detection models. With 32.98M parameters and 19.52 GFLOPs, MNDAF-Net attains the highest accuracy (F1 = 95.07% and IoU = 90.24%) of all methods. For example,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

MNDAF-Net yields 24.7% higher IoU and 11.1 points higher F1 than BIT, although BIT has far fewer parameters (3.50M). Relative to IFNet, MNDAF-Net produces a remarkable 52.9% gain in IoU and ∼28% higher F1, while using only about half the GFLOPs and achieving slightly faster inference. Similarly, compared with ChangeFormer, MNDAF-Net achieves 10.5% higher IoU and 6.2 points higher F1, but with drastically lower compute: only 19.52 GFLOPs and 62.3% shorter inference time. Through the integration of multiscale neighborhood feature decoupling and adaptive attribute sensing, MNDAF-Net achieves an ideal balance between performance and computational resource efficiency.

## VI. Conclusion

In this article, we propose MNDAF-Net, a dual-temporal remote sensing scene change detection network designed to address key challenges in fine-grained attribute change understanding. The network integrates a multiscale neighborhood decoupling mechanism with an AC strategy to jointly model local spatial dependencies and global semantic context. This framework enables precise feature separation and effective semantic filtering, supporting more accurate detection of subtle changes across complex scenes. The incorporation of the $MA^2$ and $PA^2S$ modules significantly improves the model's sensitivity to spatial structure variations and its adaptability across diverse environments. Meanwhile, the AFF-Decoder and cross-attention mechanisms enhance bitemporal feature interaction by suppressing redundant responses and emphasizing informative regions, contributing to more robust and interpretable change detection. Experimental results on three benchmark datasets demonstrate that MNDAF-Net consistently outperforms existing methods, achieving leading performance across multiple evaluation metrics. These results validate the effectiveness of the proposed approach and highlight its potential as a foundation for future developments in scene-level change analysis.

Despite its strong performance, the current model is limited to optical imagery and does not incorporate additional data modalities such as SAR or LiDAR. Furthermore, additional validation is needed in scenarios involving fine-grained category transitions or irregular temporal intervals. These aspects represent important directions for future research.

## References


[1] M. Haliem, V. Aggarwal, and B. Bhargava, "AdaPool: A diurnal-adaptive fleet management framework using model-free deep reinforcement learning and change point detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 2471–2481, Mar. 2022.

[2] G. Labadie et al., "The umbrella value of caribou management strategies for biodiversity conservation in boreal forests under global change," *Sci. Total Environ.*, vol. 907, Jan. 2024, Art. no. 168087.

[3] X. Ma, J. Yang, H. Sun, Z. Hu, and L. Wei, "Double-space environmental change detection and response strategy for dynamic multi-objective optimize problem," *Swarm Evol. Comput.*, vol. 85, Mar. 2024, Art. no. 101468.

[4] C. Wang, D. Zhao, X. Qi, Z. Liu, and Z. Shi, "A hierarchical decoder architecture for multilevel fine-grained disaster detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[5] B. Shi, Z. Jia, J. Yang, and N. K. Kasabov, "Unsupervised change detection in wide-field video images under low illumination," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1564–1576, Apr. 2023.

[6] C. Mulverhill, N. C. Coops, and A. Achim, "Continuous monitoring and sub-annual change detection in high-latitude forests using harmonized Landsat Sentinel-2 data," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 309–319, Mar. 2023.

[7] X. Yu and X. Yue, "Similarity matrix entropy for multitemporal polarimetric SAR change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[8] C. Wu, H. Chen, B. Du, and L. Zhang, "Unsupervised change detection in multitemporal VHR images based on deep kernel PCA convolutional mapping network," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 12084–12098, Nov. 2022.

[9] W. Xiao, X. Huang, F. He, J. Silva, S. Emrani, and A. Chaudhuri, "Online robust principal component analysis with change point detection," *IEEE Trans. Multimedia*, vol. 22, no. 1, pp. 59–68, Jan. 2020.

[10] J. Li, J. Huang, L. Zheng, and W. Zheng, "Study on a mother wavelet optimization framework based on change-point detection of hydrological time series," *Hydrol. Earth Syst. Sci.*, vol. 27, no. 12, pp. 2325–2339, Jun. 2023.

[11] L. Ru, B. Du, and C. Wu, "Multi-temporal scene classification and scene change detection with correlation based fusion," *IEEE Trans. Image Process.*, vol. 30, pp. 1382–1394, 2021.

[12] Y. Peng, B. Cui, H. Yin, Y. Zhang, and P. Du, "Automatic SAR change detection based on visual saliency and multi-hierarchical fuzzy clustering," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7755–7769, 2022.

[13] B. Cui et al., "Enhanced edge information and prototype constrained clustering for SAR change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[14] G. Shi, "Adaptive multisensor fusion for remote sensing change detection using USASE," *IEEE Sensors J.*, vol. 25, no. 7, pp. 12265–12277, Apr. 2025.

[15] D. Li, S. Yan, M. Zhao, and T. W. S. Chow, "Spatiotemporal tree filtering for enhancing image change detection," *IEEE Trans. Image Process.*, vol. 29, pp. 8805–8820, 2020.

[16] M. Zhang, Q. Li, Y. Yuan, and Q. Wang, "Boosting binary object change detection via unpaired image prototypes contrast," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[17] L. Bouchelkia, A. Tahraoui, and R. Kheddam, "Hybrid model CNN-SVM for land change detection," in *Proc. 2nd Int. Conf. Electr. Eng. Autom. Control (ICEEAC)*, Setif, Algeria, May 12, 2024, pp. 1–6.

[18] M. Wang, X. Li, K. Tan, J. Mango, C. Pan, and D. Zhang, "Position-aware graph-CNN fusion network: An integrated approach combining geospatial information and graph attention network for multiclass change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[19] L. Wang, J. Zhang, and L. Bruzzone, "MixCDNet: A lightweight change detection network mixing features across CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[20] W. Li, L. Xue, X. Wang, and G. Li, "ConvTransNet: A CNN–transformer network for change detection with multiscale global–local representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[21] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[22] B. Jiang et al., "VcT: Visual change transformer for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[23] Y. Cui, H. Chen, S. Dong, G. Wang, and Y. Zhuang, "U-shaped CNN-ViT Siamese network with learnable mask guidance for remote sensing building change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11402–11418, 2024.

[24] L. Song et al., "Multi-granularity Siamese transformer-based change detection in remote sensing imagery," *Eng. Appl. Artif. Intell.*, vol. 136, Oct. 2024, Art. no. 108960.

[25] Z. Lv, M. Zhang, W. Sun, J. A. Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[26] Y. Xu et al., "Remote sensing change detection method based on dynamic adaptive focal loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[27] J. Pan, Y. Bai, Q. Shu, Z. Zhang, J. Hu, and M. Wang, "M-Swin: Transformer-based multiscale feature fusion change detection network within cropland for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.

[28] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[29] R. Song, Y. Feng, C. Xing, Z. Mu, and X. Wang, "Hyperspectral image change detection based on active convolutional neural network and spatial–spectral affinity graph learning," *Appl. Soft Comput.*, vol. 125, Aug. 2022, Art. no. 109130.

[30] Z. Dong et al., "Mapping inundation extents in Poyang lake area using Sentinel-1 data and transformer-based change detection method," *J. Hydrol.*, vol. 620, May 2023, Art. no. 129455.

[31] T. Zhu et al., "FTA-Net: Frequency-temporal-aware network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3448–3460, 2025.

[32] J. Wang et al., "MDANet: A multi-stage domain adaptation framework for generalizable low-light image enhancement," *Neurocomputing*, vol. 627, Apr. 2025, Art. no. 129572.

[33] P. Chen, B. Zhang, D. Hong, Z. Chen, X. Yang, and B. Li, "FCCDN: Feature constraint network for VHR image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 101–119, May 2022.

[34] S. Tian, X. Tan, A. Ma, Z. Zheng, L. Zhang, and Y. Zhong, "Temporal-agnostic change region proposal for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 306–320, Oct. 2023.

[35] W. Wang, C. Liu, G. Liu, and X. Wang, "CF-GCN: Graph convolutional network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[36] Y. Shangguan, J. Li, Z. Chen, L. Ren, and Z. Hua, "Multiscale attention fusion graph network for remote sensing building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[37] X. Zhang et al., "VGGM: Variational graph Gaussian mixture model for unsupervised change point detection in dynamic networks," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 4272–4284, 2024.

[38] Z. Zhang, Z. Geng, and Y. Han, "Graph structure change-based anomaly detection in multivariate time series of industrial processes," *IEEE Trans. Ind. Informat.*, vol. 20, no. 4, pp. 6457–6466, Apr. 2024.

[39] J. Miao, S. Li, X. Bai, W. Gan, J. Wu, and X. Li, "RS-NormGAN: Enhancing change detection of multi-temporal optical remote sensing images through effective radiometric normalization," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 324–346, Mar. 2025.

[40] F. Xiong, T. Li, Y. Yang, J. Zhou, J. Lu, and Y. Qian, "Wavelet Siamese network with semi-supervised domain adaptation for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[41] L. Wang, Z.-H. You, W. Lu, S.-B. Chen, J. Tang, and B. Luo, "Attention-aware Sobel graph convolutional network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[42] Y. Zhang, X. Song, Z. Hua, and J. Li, "CGMMA: CNN-GNN multiscale mixed attention network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7089–7103, 2024.

[43] F. Yang, Y. Yuan, A. Qin, Y. Zhao, T. Song, and C. Gao, "Change-aware cascaded dual-decoder network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[44] C. Zhang, L. Wang, and S. Cheng, "HCGNet: A hybrid change detection network based on CNN and GNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[45] J. Long, M. Li, X. Wang, and A. Stein, "Semantic change detection using a hierarchical semantic graph interaction network from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 318–335, May 2024.

[46] K. Tang, F. Xu, X. Chen, Q. Dong, Y. Yuan, and J. Chen, "The ClearSCD model: Comprehensively leveraging semantics and change relationships for semantic change detection in high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 299–317, May 2024.

[47] W. Cheng, Y. Feng, L. Song, and X. Wang, "DMF2Net: Dynamic multi-level feature fusion network for heterogeneous remote sensing image change detection," *Knowl.-Based Syst.*, vol. 300, Sep. 2024, Art. no. 112159.

[48] H. Chen, J. Song, C. Wu, B. Du, and N. Yokoya, "Exchange means change: An unsupervised single-temporal change detection framework based on intra- and inter-image patch exchange," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 87–105, Dec. 2023.

[49] R. Zhang, H. Zhang, X. Ning, X. Huang, J. Wang, and W. Cui, "Global-aware Siamese network for change detection on remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 61–72, May 2023.

[50] Y. Sun, L. Lei, Z. Li, and G. Kuang, "Similarity and dissimilarity relationships based graphs for multimodal change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 70–88, Feb. 2024.

[51] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[52] X. Xu, Z. Yang, and J. Li, "AMCA: Attention-guided multiscale context aggregation network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[53] X. Tang, T. Zhang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "WNet: W-Shaped hierarchical network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[54] B. Fang, G. Chen, R. Kou, M. E. Paoletti, J. M. Haut, and A. Plaza, "CIT: content-invariant translation with hybrid attention mechanism for unsupervised change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 321–339, Oct. 2023.

[55] W. Yu, L. Zhuo, and J. Li, "GCFormer: Global context-aware transformer for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[56] Z. Li et al., "STADE-CDNet: Spatial–temporal attention with difference enhancement-based network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[57] M. Jiang, Y. Chen, Z. Dong, X. Liu, X. Zhang, and H. Zhang, "Multiscale fusion CNN-transformer network for high-resolution remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5280–5293, 2024.

[58] T. Lei et al., "Lightweight structure-aware transformer network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[59] J. Liu, S. Li, R. Dian, Z. Song, and X. Kang, "MDENet: Multidomain differential excavating network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[60] M. Hu, C. Wu, and L. Zhang, "GlobalMind: Global multi-head interactive self-attention network for hyperspectral change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 465–483, May 2024.

[61] S. Qian, Z. Xue, M. Jia, Y. Chen, and H. Su, "Temporal-spectral-semantic-aware convolutional transformer network for multi-class tidal wetland change detection in greater bay area," *ISPRS J. Photogramm. Remote Sens.*, vol. 216, pp. 126–141, Oct. 2024.

[62] A. Hassani and H. Shi, "Dilated neighborhood attention transformer," 2022, *arXiv:2209.15001*.

[63] J. Li, H. Zhang, R. Song, W. Xie, Y. Li, and Q. Du, "Structure-guided feature transform hybrid residual network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[64] Y. Li, F. Wei, Y. Zhang, W. Chen, and J. Ma, "HS2P: Hierarchical spectral and structure-preserving fusion network for multimodal remote sensing image cloud and shadow removal," *Inf. Fusion*, vol. 94, pp. 215–228, Jun. 2023.

[65] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[66] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[67] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[68] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[69] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[70] Z. Wang, G. Gu, M. Xia, L. Weng, and K. Hu, "Bitemporal attention sharing network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 10368–10379, 2024.

[71] Z. Zhan, H. Ren, M. Xia, H. Lin, X. Wang, and X. Li, "AMFNet: Attention-guided multi-scale fusion network for bi-temporal change detection in remote sensing images," *Remote Sens.*, vol. 16, no. 10, p. 1765, May 2024.

[72] Z.-H. You, S.-B. Chen, J.-X. Wang, and B. Luo, "Robust feature aggregation network for lightweight and effective remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 215, pp. 31–43, Sep. 2024.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.



**Zhengpeng Li** (Graduate Student Member, IEEE) received the B.S. degree in engineering and the master's degree from the University of Science and Technology Liaoning, Anshan, China, in 2016 and 2020, respectively, where he is currently pursuing the Ph.D. degree with a focus on remote sensing image processing and deep learning.

In 2020, he studied at the University of Science and Technology Liaoning, with research directions of deep learning, remote sensing scene understanding, and remote sensing semantic image segmentation.

Mr. Li is a member of GRSS. He is also a reviewer for journals such as *Signal, Image and Video Processing*, *Multimedia Systems*, *International Journal of Machine Learning and Cybernetics*, *International Journal of Multimedia Information Retrieval*, *Journal of Real-Time Image Processing*, *The Visual Computer*, *The Journal of Supercomputing*, and *Scientific Reports*.



**Jun Hu** (Member, IEEE) received the B.S. degree in hydraulic engineering from North China University of Water Conservancy and Electric Power (now North China University of Water Resources and Electric Power), Zhengzhou, China, in 2001, and the Ph.D. degree in disaster prevention and mitigation engineering and protective engineering from Dalian University of Technology, Dalian, China, in 2008, through a direct bachelor's-to-Ph.D. program.

In November 2014, he was exceptionally promoted to a Professor. He is currently the President of the University of Science and Technology Liaoning, Anshan, China. His research interests are focused on remote sensing imagery, geotechnical engineering numerical simulation, and the automation of safety monitoring and early warning systems for tailings dams. He has led over 30 key projects sponsored by the Education Department of Liaoning Province, general projects, and corporate commissioned projects, with total funding exceeding 30 million yuan. As the first author or the corresponding author, he has published more than 60 articles in international and domestic journals.

Dr. Hu has been awarded the title of the Outstanding Teacher of Liaoning Province, among other honors.



**Bin Yang** received the bachelor's degree in civil engineering from the University of Science and Technology Liaoning, Anshan, China, in 2013, and the master's and Ph.D. degrees in geotechnical engineering from Northeastern University, Shenyang, China, in 2015 and 2019, respectively.

He is currently an Associate Professor and a Master's Supervisor with the School of Civil Engineering, University of Science and Technology Liaoning. He has led two provincial-level projects, one university-level project, and four enterprise-commissioned projects, securing over 2.6 million RMB in research funding. He has published more than 30 academic articles and co-authored a monograph, which was funded by the National Publishing Fund. He holds one utility model patent and two software copyrights. In addition, he has led one provincial-level graduate education reform project, one Ministry of Education industry–university collaborative education project, and one university-level education reform project, with three published articles on educational reform.

Dr. Yang received the "Ten Thousand Talents" Award from Liaoning Province Hundred-Thousand-Talents Program.



**Kunyang Wu** received the B.S. degree in measuring and control technology and instrumentation from Jilin University, Changchun, China, in 2021, where he is currently pursuing the Ph.D. degree in detection technique and automatic device.

His research interests focus on the field of detection techniques and automatic devices, with a specific interest in multimodal sensor calibration, sensor fusion, and the utilization of perception systems for autonomous navigation and vision applications. By exploring these domains, he aims to contribute to the advancement of cutting-edge technologies in autonomous systems.



**Jiansheng Wu** received the master's degree in computer application technology from Anshan University of Science and Technology (now University of Science and Technology Liaoning), Anshan, China, in 2005.

He was promoted to a Professor in 2010. He is currently a Faculty Member with the School of Computer and Software Engineering, University of Science and Technology Liaoning. His research interests focus on computer vision, image processing and analysis, multimodal remote sensing image analysis, and medical image analysis. He has led or participated in projects funded by the National Natural Science Foundation of China, key projects of the Education Department of Liaoning Province, general projects, and corporate-commissioned projects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:37:53 UTC from IEEE Xplore. Restrictions apply.